

1. Create variables of all Python primitive data types. Print their:

- Value
- Data Type
- Memory Address (id())

2. Create three variables

- a = 15
- b = 4

- Addition
- Subtraction
- Multiplication
- Division
- Floor Division
- Modulus
- Exponent

3. Create two variables:

- x = "100"
- y = 50

- Convert the variables so that you can perform arithmetic operations between them.

4. Predict the output **before running the program**, then verify.

```
a = 10
b = 20

print(a + b * 2)
print((a + b) * 2)
print(a ** 2 + b)
print(a // 3)
print(a / 3)
```

5. Write a program to calculate electricity bill.

Use variables for:

- Units Consumed
- Cost per Unit
- Calculate total bill and GST (18%).

6. Create a list of programming languages.

- Check whether:

- "Python" exists.
- "Java" does not exist.
- "C++" exists.
- Use Membership Operators.

7. Write a program that converts:

- int $\rightarrow$ float
- float $\rightarrow$ string
- string $\rightarrow$ int
- int $\rightarrow$ bool
- bool $\rightarrow$ int

8. Predict the output.

```
a = 10
a += 5
a *= 2
a -= 8
a /= 4

print(a)
```

9. Predict the output.

```
print(5 == 5.0)
print(5 is 5.0)
```

10. Create a program that stores product information.

Store:

- Product Name
- Price
- Quantity

Calculate:

- Total Cost
- GST (18%)
- Final Amount

11. Write a program that checks whether a person can register for a driving license.

Conditions:

- Age ≥ 18
- Indian Citizen

12. Predict the output.

```
a = [10,20]
b = a
c = [10,20]

print(a == b)
print(a is b)

print(a == c)
print(a is c)

print(id(a))
print(id(b))
print(id(c))
```